Waze Churn Rate Analysis

Google Advanced Data Analyst Course 1 Project - Garrett Hanson

Project Overview

The Project is making a predictive model that will accurately predict churn rate, and guide stakeholders towards making informed decisions that can decrease churn rate. I am the newest member of the Waze data analyst team and have been tasked with preparing the data.

Key Insights

- 'label' column is missing 700 values.
- Most columns are int or float Dtypes, but 'label' and 'device' are object Dtypes.
- There appears to be no significant difference between the rows with and without null values in 'label'
- 17.7% of users are labeled as 'churned'
- Users that have churned appear to users that drove significantly more per day and had more drives per day.

Details

```
In [24]: # Add a column to df called `km_per_driving_day`
         df['km_per_driving_day'] = df['driven_km_drives'] / df['driving_days']

         # Group by `label`, calculate the median, and isolate for km per driving day
         df.groupby('label')['km_per_driving_day'].median(numeric_only=True)

Out[24]: label
         churned     697.541999
         retained    289.549333
         Name: km_per_driving_day, dtype: float64
```

Now, calculate the median number of drives per driving day for each group.

```
In [25]: # Add a column to df called `drives_per_driving_day`
         df['drives_per_driving_day'] = df['drives'] / df['driving_days']

         # Group by `label`, calculate the median, and isolate for drives per driving day
         df.groupby('label')['drives_per_driving_day'].median(numeric_only=True)

Out[25]: label
         churned     10.0000
         retained     4.0625
         Name: drives_per_driving_day, dtype: float64
```

Image showing how churned users drive more KM per day and do more drives per day.

Next Steps

The data team should inquire about how an when a user is classified as 'churned' in the data set.

The team should do research into why the 'label' column is missing values.

The team should require if this dataset is representative of all users.

The team should do research into if the data is logically consistent.